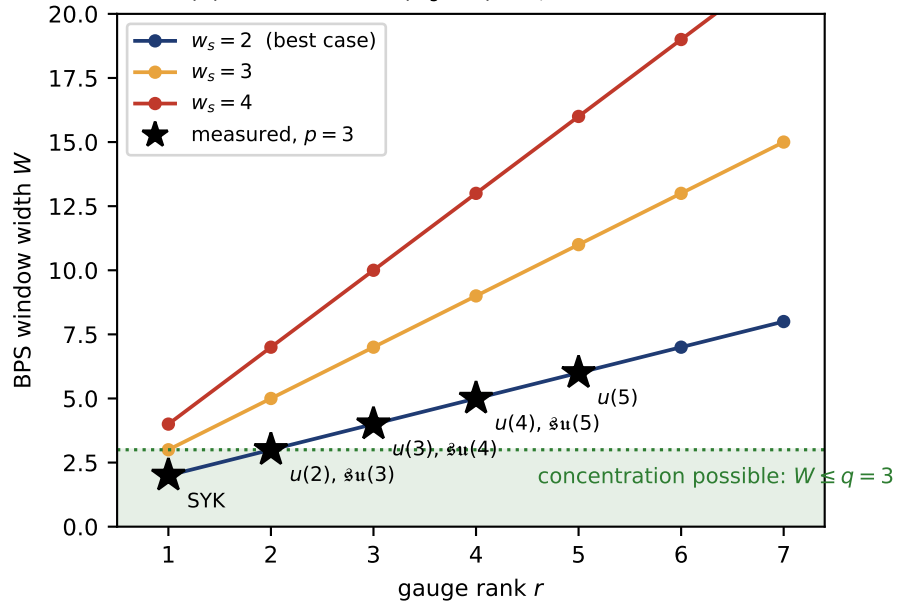


(a) one law: $W = r(w_s - 1) + 1$, with SYK at $r = 1$



(b) w_s cycles $q-1, q, q+1, q$ with period 4 in p

